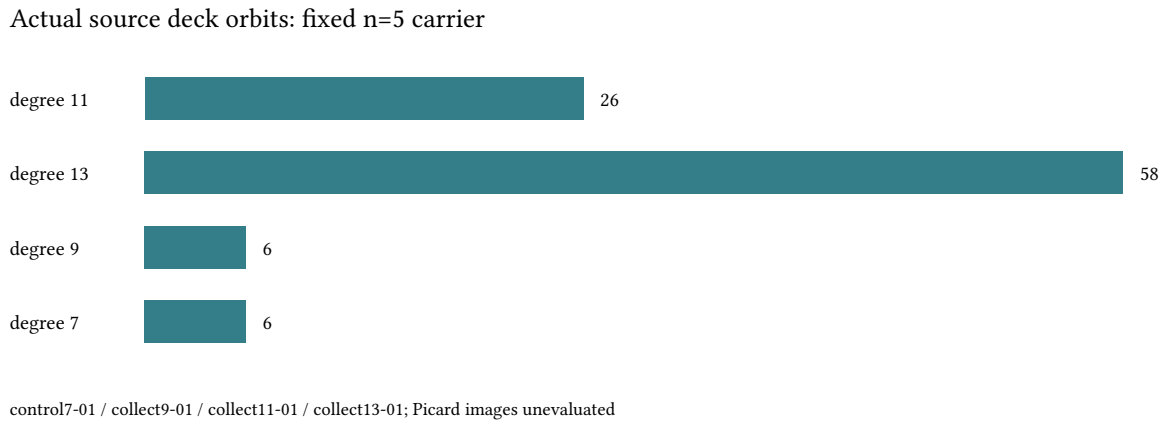
R45: larger actual-source collector

Observations and conditional additive-group proofs. No independent scientific verdict, Picard-image claim or IC efficiency comparison.



What was enumerated

The R39 native collector was widened to 16-bit coordinates and 32-bit polynomial convolution. It enumerates the same fixed ordinary chart, lambda choices and physical coefficient descent. Every accepted source point passes all five original equations, eight registered units, exact Tate order 11 and the local rank-two check. All deck and coefficient Frobenius images are rechecked. The degree-7 control reproduces the complete old point export byte for byte. The accepted cases are 420/540/2860/7540 rational points, respectively, 60/60/260/580 closed points and 6/6/26/58 deck orbits for degrees 7/9/11/13. Four native attempts and one Lean workflow pass. Three named theorems and one generated equation pass the audit, with zero failing or trusted declarations. The exact complete point records are in results.json and `runs/*/points/closed-points.txt`. All attempts, caps and exits are retained.

Degrees 9 and 11 were frozen before data. Degree 13 was added by a prospective amendment after their outcomes, with unchanged 4 GiB / 1200 s build/run caps. All field products agree with an independent carryless polynomial reduction; all nonzero inverses and every quadratic inverse pass. Basis triples check field identities; the old cubic exhaustive loop is not repeated at larger sizes. Composite degree 9 is classified by actual tuple periods, including any subfield contributions rather than assuming every point has degree 9.

Corrected search priority

R13 already exhausted the ordinary degree-500 and degree-580 supports and proved their retained mixed images vanish by full scalar-fiber averaging, under its written modular/pairing bridge. R44's recommendation to screen them again was stale. The sign obstruction in R44 is a second rejection mechanism. R45 instead collects new source supports with ten distinct deck/Frobenius classes, which avoid these immediate stabilizer cancellations.

Formal scope and next obligations

The Lean module proves zero mixed projection from sign fixation and from scalar two-step fixation, and proves that a nonzero projection requires both to fail. These are conditional additive-group statements. They do not import native arrays, formalize global geometry, or show that the new classes are nonprincipal. Native compile and axiom-audit receipts are in results.json.

The next calculation is complete divisor evaluation and signed Picard classes on the new supports, followed by actual target images and dependency checks. Proper level-77 chosen-model transfer, quotient saturation, polarization, theta descent and node187 coupled divisors remain separate obligations. Larger residue fields on this fixed n=5 carrier are not a larger-n Koblitz transfer. Independent scientific replay and semantic review remain pending.

Sources: R13 REPORT/PROOF_MAP, R39 collector, R41 signed classes, R44 orientation; primary model and etale references with scope boundaries in LITERATURE.md.